

Data-Driven Pharmaceutical Sales Planning & Doctor Targeting Model

A Multi-Attribute Decision Model for Resource Allocation
under Constraint in a Regional Product Launch

Complete Strategy Documentation with Formula Derivations,
Justifications and Academic References

Scope of the problem	600 doctors · 6 medical representatives · ₹30 lakh quarterly budget · 6-month horizon
Core instrument	Doctor Value Score (DVS) — a five-pillar weighted additive model
Methodological basis	Multi-Attribute Utility Theory; sales-response optimisation; diffusion theory
Formulas derived	10, each with proof, assumptions, limitations and worked example
References cited	23 academic, industry and regulatory sources
Supporting artefact	DVS Manager — a working web application implementing the model

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HOW TO READ THIS REPORT

Sections 1–4 set out the business problem and the design of the scoring instrument. **Sections 5–10 are the analytical core:** each presents one formula, derives it from first principles, states the theoretical authority behind it, lists the assumptions required for validity, acknowledges its limitations, and verifies it against a worked example. Section 11 covers execution, Section 12 offers a critical self-evaluation, and the References map every source to the specific formula it supports.

1 Executive Summary

The problem, the instrument, and what the model delivers

This report documents a complete, quantitatively grounded strategy for launching a new pharmaceutical molecule in a regional market under severe resource constraint. It is presented as an academic exercise in **applied decision science**: the objective is not merely to propose a commercial plan, but to demonstrate that every element of that plan follows from a defensible model, that each formula used is derived rather than asserted, and that the assumptions on which it rests are stated openly and tested.

The problem. A pharmaceutical company must establish a new molecule in a region containing approximately 600 relevant prescribers. It has six medical representatives (MRs), a promotional budget of ₹30 lakhs per quarter, and roughly six months in which the launch will succeed or fail. Simple arithmetic shows that uniform coverage is not viable. Six representatives making 8–10 calls a day across 22 working days generate roughly 1,200 calls per month; spread across 600 doctors that is one visit per doctor every two months — too infrequent to build a prescribing relationship. Spreading ₹30 lakhs evenly gives ₹5,000 per doctor per quarter, an amount too small to be meaningful to anyone. The constraint is therefore not a matter of efficiency but of **strategic selection**.

The instrument. The strategy replaces uniform coverage with a composite scoring model. Every doctor is evaluated on five weighted dimensions — Prescription Potential (30%), Adoption Propensity (25%), Influence and Network Power (20%), Accessibility (15%) and Competitive Vulnerability (10%) — producing a single **Doctor Value Score (DVS)** on a 0–10 scale. Doctors are then banded into four tiers, each receiving a differentiated call frequency and budget entitlement.

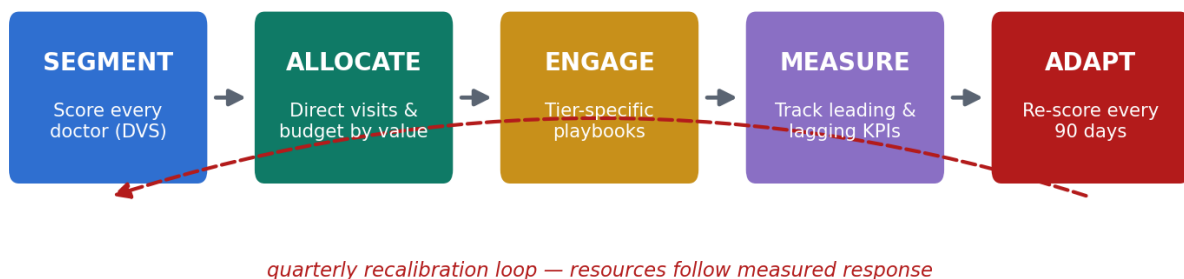


Figure 1.1 — The five-stage strategic cycle. Segmentation feeds allocation, which drives engagement; measurement closes the loop through quarterly recalibration.

Why this is defensible. The scoring formula is not an arbitrary weighting exercise. It is the weighted additive value function of Multi-Attribute Utility Theory (Keeney & Raiffa, 1976), the canonical method for combining multiple criteria into a single preference ordering. The differentiated call frequencies follow from the empirical finding that sales response to detailing is concave (Lodish, 1971; Manchanda et al., 2004): under a binding capacity constraint, the optimal allocation equalises the marginal return per call across doctors, and a tiered frequency ladder is a discrete approximation of exactly that optimum. Each of these results is derived in full in Sections 5 and 7.

Headline outcomes of the model

Dimension	Uniform coverage	Value-based model	Basis
Spend per high-value doctor	₹5,000/quarter	₹20,770/quarter	Formula 5 — \$8
Visit frequency, top tier	0.5 per month	3.5 per month	Formula 4 — \$7
Doctors driving ~75% of revenue	undifferentiated	200 identified	Formula 2 — \$6
Field-force utilisation	not modelled	68% (feasible)	Formula 3 — \$7
Adaptation cycle	annual review	90 days	\$11.3

THE CENTRAL ARGUMENT IN ONE SENTENCE

Because promotional response is concave and field capacity is finite, spreading effort evenly across 600 doctors is provably sub-optimal — and a weighted, multi-dimensional score is the most defensible available basis on which to concentrate that effort.

2 The Business Problem

Why the conventional approach fails, stated quantitatively

2.1 The constraint set

Four constraints operate simultaneously. Each on its own would be manageable; together they make uniform coverage arithmetically impossible.

Constraint	Value	Implication
Field force	6 medical representatives	A scarce, expensive asset. Every call must be justified by expected return.
Doctor universe	500–800 relevant prescribers	Physically cannot all be covered. Selection is not optional — it is forced.
Promotional budget	₹30 lakhs per quarter	Every rupee spent on a low-value doctor is a rupee withheld from a high-value one.
Time	6 months to establish	First-mover prescribing habits form early; a wait-and-see approach forfeits the window.

2.2 The arithmetic of failure

The case against uniform coverage does not rest on judgement. It follows from the numbers.

Coverage failure

$$\text{Visits per doctor per month} = (6 \text{ MRs} \times 22 \text{ days} \times 9 \text{ calls}) \div 600 \text{ doctors} = 1.98$$

Under equal division each doctor receives roughly two calls per month only if every call is spent on doctors — leaving nothing for chemists, hospitals or administration. Reserving realistic non-doctor time reduces this to approximately one visit every two months.

Pharmaceutical detailing depends on repetition and relationship. A contact frequency of once per two months is widely regarded as below the threshold at which brand recall and prescribing habit form. The coverage is nominal rather than effective.

Budget dilution

$$₹30,00,000 \div 600 \text{ doctors} = ₹5,000 \text{ per doctor per quarter}$$

At this level the allocation funds neither meaningful sampling, nor a CME programme, nor scientific engagement of any substance. It is spread too thinly to change behaviour anywhere.

Absence of prioritisation

Under uniform coverage a representative spends the same time with a medical-college professor who teaches forty residents as with a general practitioner who will never prescribe in the therapy area. The opportunity cost is invisible because it is never measured.

Absence of adaptation

A plan fixed at Month 1 and reviewed annually cannot respond to the information generated by the launch itself. Doctors who respond early and doctors who do not respond at all receive identical treatment for twelve months.

THE REFRAMED QUESTION

The strategic question is not “how do we reach all 600 doctors?” — a question whose honest answer is “we cannot”. It is **“which 200 doctors will generate the majority of our revenue, and how do we secure their prescribing behaviour?”** This is the Pareto principle (Pareto, 1896) applied with measurement rather than intuition.

2.3 What a solution must satisfy

Any acceptable model must meet five conditions, which together define the design brief for the remainder of this report:

- **Multi-dimensional.** Ranking doctors by patient volume alone is demonstrably insufficient — a high-volume doctor loyal to a competitor is worth less than a medium-volume early adopter.
- **Feasible.** The resulting call plan must fit inside real field capacity, and this must be demonstrated rather than assumed.
- **Proportional.** Budget and visit frequency must track expected value, not headcount.
- **Adaptive.** The model must re-score doctors on observed behaviour at a defined interval.
- **Transparent.** Every number must be traceable to a formula, and every formula to a stated justification — otherwise the model cannot be audited or defended.

3 Strategic Framework and Guiding Principles

The architecture within which the formulas operate

3.1 Five stages

The strategy is organised as a closed loop of five stages. The first three convert data into action; the last two convert outcomes back into data, which is what makes the system adaptive rather than static.

Stage	Function	Governing formula
Segment	Score and classify every doctor on five dimensions	F1 (DVS), F2 (tiering)
Allocate	Direct visit frequency and budget in proportion to value	F3, F4 (capacity), F5 (budget)
Engage	Deploy tier-specific engagement playbooks	§11.1 (qualitative)
Measure	Track leading and lagging indicators	§11.2 (KPI set)
Adapt	Re-score on observed behaviour every 90 days	F1 re-applied; §11.3

3.2 Guiding principles

Five principles govern design decisions throughout. Each is stated so it can be argued against — a principle that cannot be contradicted is not a principle but a platitude.

Principle 1 — Concentration over dispersion

It is better to influence 200 doctors deeply than to touch 600 superficially. This follows directly from concavity of promotional response (Section 7.3): if response were linear, dispersion and concentration would be equivalent, and this principle would be false.

Principle 2 — Value is what a doctor can deliver, not who they are

Seniority, reputation and title are proxies, and often poor ones. The model scores expected contribution, which is why a high-profile doctor locked into a competitor brand can rank below a less prominent early adopter.

Principle 3 — Channel follows accessibility

A doctor who scores highly on value but very low on accessibility should not be assigned to routine representative calls. The correct response is to change the channel — senior-manager contact, peer-to-peer engagement, or digital — rather than to abandon the opportunity or waste calls on it.

Principle 4 — Continuous recalibration

The model re-scores every doctor every 90 days on observed prescribing and engagement. Initial scores are a structured prior; behaviour is evidence, and evidence supersedes the prior.

Principle 5 — Opinion leaders are force multipliers, with caveats

Key Opinion Leaders influence prescribing beyond their own clinic — a finding established in the classic Columbia tetracycline study (Coleman, Katz & Menzel, 1966). However, that same literature contains a serious warning about over-claiming this effect, which is examined honestly in Section 12.2.

3.3 How the principles constrain the formulas

These principles are not decorative. Each one rules out a design that would otherwise be available, and it is worth stating those exclusions explicitly:

Principle	What it rules out	Where it binds
Concentration over dispersion	Equal call frequency across all tiers	Formula 4 (\$7.3)
Value, not status	Ranking by seniority, decile or patient volume alone	Formula 1 (\$5)
Channel follows accessibility	Disqualifying inaccessible doctors outright	Pillar 4 (\$4.3)
Continuous recalibration	A fixed annual plan	\$11.3
Multipliers with caveats	Claiming unattributable KOL influence as measured	\$12.2

The relationship runs in both directions. The principles justify the formulas, and the formulas make the principles operational — a principle that cannot be expressed as a decision rule has no effect on what a representative does on Monday morning.

A NOTE ON INTELLECTUAL HONESTY

A model of this kind is only as credible as its treatment of its own weaknesses. Section 12 is therefore devoted to a critical evaluation: where the assumptions are strong, where the evidence is thin, and which claims should be presented as modelled rather than measured. This is included deliberately — a strategy document that reports only its strengths should not be trusted with a ₹87 lakh decision.

4 The Five-Pillar Doctor Value Score

What is measured, how it is scored, and why each dimension is included

4.1 Why a multi-dimensional model

Conventional pharmaceutical segmentation ranks doctors on a single axis: patient volume, usually expressed as deciles of prescribing in the therapy area. This is convenient and dangerously incomplete.

THE MOTIVATING COUNTER-EXAMPLE

Consider two doctors. **Doctor A** sees 50 relevant patients a day but is loyal to a competitor brand, sits on that competitor’s advisory board, and has adopted no new molecule in four years. **Doctor B** sees 20 relevant patients a day, has adopted three new molecules in two years, teaches at a medical college, and has voiced dissatisfaction with existing therapy. A volume-ranked model places Doctor A far above Doctor B. Every practical consideration says the opposite. A single-axis model cannot represent this; a multi-attribute model can.

The model therefore evaluates five dimensions simultaneously. The weights are shown below and justified individually in the pillar descriptions that follow; the formal justification for combining them additively is given in Section 5.

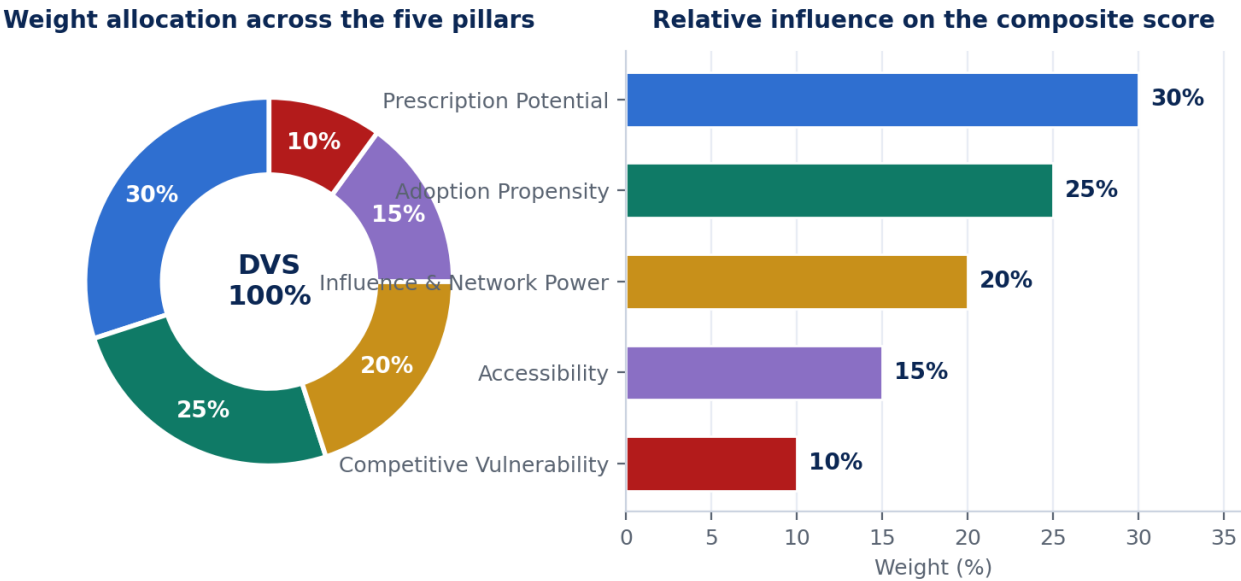


Figure 4.1 — Pillar weights. Prescription Potential dominates because it maps directly onto prescriptions; the remaining pillars modify the probability that this potential is realised.

4.2 Weight rationale

Weights were assigned by direct allocation, the method Edwards (1977) terms SMART. The ordering follows a single logic: **a pillar’s weight reflects how directly it determines realised prescriptions, discounted by how reliably it can be measured.**

Pillar	Weight	Why this weight
Prescription Potential	30%	Only pillar mapping directly to the outcome variable; also the most reliably measurable.
Adoption Propensity	25%	Determines whether potential is realised within the launch window. Critical under time constraint.

Pillar	Weight	Why this weight
Influence & Network Power	20%	Extends effect beyond the individual, but the multiplier is indirect and hard to attribute.
Accessibility	15%	Converts theoretical into achievable value; moderate weight because low scores should change channel, not disqualify.
Competitive Vulnerability	10%	Genuinely relevant but the most volatile and least reliably observed variable.

AN HONEST STATEMENT ABOUT THE WEIGHTS

These weights are **judgemental, not statistically estimated**. They encode expert reasoning about the therapy area, not a regression fitted to prescription data — because at launch no such data exists for a new molecule. They should be treated as a structured prior. Once two to three quarters of actual prescribing data are available, the correct procedure is to regress observed prescriptions on the five pillar scores and re-estimate the weights empirically. Section 12.1 develops this point.

4.3 The pillars in detail

Each pillar is presented with its guiding question, scoring rubric, data requirements, sources, and the reasoning behind its inclusion and weight.

Pillar 1 · Prescription Potential		Weight 30%
“How many relevant prescriptions can this doctor generate for our molecule?”		
Score	Descriptor	
10	80+ relevant patients per month	
8	50–79 relevant patients	
6	30–49 relevant patients	
4	15–29 relevant patients	
2	Below 15 relevant patients	

Data collected: Average patients seen per day; Percentage presenting with the target condition; Prescriptions written in the therapy area per month; Seasonal variation in footfall; Practice type (solo clinic, multi-specialty, hospital).

Sources: IQVIA patient-volume estimates; MR field assessment during pre-launch mapping; Hospital OPD records; Local chemist prescription-pattern data.

Why this pillar, and why this weight. This pillar carries the highest weight because it is the only one that maps directly onto the dependent variable: prescriptions. The others modify the probability that this potential is realised. **Critically, volume must be relevant volume** — a dermatologist seeing 100 patients a day scores zero for a cardiology molecule. Scoring total footfall rather than target-condition footfall is the single most common error in pharmaceutical targeting.

Pillar 2 · Adoption Propensity**Weight
25%**

“How likely is this doctor to actually prescribe a NEW molecule?”

Score	Descriptor
10	Early adopter — tries new molecules within 3 months of launch
7	Early majority — adopts after peer evidence emerges (3–6 months)
4	Late majority — waits until well established (6–12 months)
2	Laggard / loyalist — almost never switches brand

Data collected: History of adopting new brands in previous launches; Current brand-loyalty concentration; Attitude toward representative visits; Participation in CMEs and conferences; Response to past sampling programmes.

Sources: IQVIA / internal CRM behaviour on prior launches; Pre-launch MR survey; Prescription-basket diversity analysis.

Why this pillar, and why this weight. This pillar is weighted second because in a time-constrained launch speed of adoption is nearly as valuable as volume. It rests on Rogers’ diffusion theory (2003), which classifies adopters as Innovators (2.5%), Early Adopters (13.5%), Early Majority (34%), Late Majority (34%) and Laggards (16%). The four-level rubric compresses these into operationally distinguishable categories. **A medium-volume early adopter outranks a high-volume laggard**, because the laggard will not prescribe within the launch window regardless of effort.

Pillar 3 · Influence & Network Power**Weight
20%**

“Does this doctor influence the prescribing behaviour of other doctors?”

Score	Descriptor
10	Regional or national KOL, medical-college professor, association office-bearer
7	Local influencer, conference speaker, mentors junior doctors
5	Moderate peer influence, respected in the local medical community
2	Solo practitioner with limited network influence

Data collected: KOL status; Teaching appointment at a medical college; Conference speaking record; Department headship or association office; Inbound referral volume from peers; Presence in medical social media.

Sources: Conference speaker rosters; Medical association office-bearer lists; Referral-pattern analysis; Medical community platforms (e.g. Docplexus).

Why this pillar, and why this weight. Influence is included as a separate dimension because a converted opinion leader affects prescribing beyond their own practice. The theoretical basis is the Columbia tetracycline study (Coleman, Katz & Menzel, 1966), which found that network-central physicians adopted earlier and that their adoption preceded the acceleration of the diffusion curve. That study also supplies a usable operational definition: opinion leaders were identified by receiving three or more sociometric nominations — that is, by measured referral and peer nomination rather than by job title. Section 12.2 sets out an important caveat on the size of this effect.

Pillar 4 · Accessibility**Weight
15%**

“How easily can this doctor be reached and meaningfully engaged?”

Score	Descriptor
10	Easily accessible, welcomes representatives, allows adequate time
7	Moderately accessible at specific times or on specific days
4	Difficult to access but reachable through alternative channels
1	Virtually inaccessible; refuses representative meetings

Data collected: Travel time and distance from the representative’s base; Availability for meetings; Preferred engagement channel; Gatekeeping barriers (hospital policy, appointments); Historical visit-to-meeting conversion rate.

Sources: MR territory mapping; CRM visit logs; Route-planning data.

Why this pillar, and why this weight. This is the efficiency dimension: it converts theoretical value into achievable value. A perfectly targeted doctor who cannot be met consumes calls without producing prescriptions. Its weight is deliberately moderate (15%) because low accessibility should trigger a **change of channel rather than abandonment** — an operational response the additive formula cannot itself express, discussed as a known limitation in Section 5.5.

Pillar 5 · Competitive Vulnerability**Weight
10%**

“How entrenched is the competition, and can we realistically break through?”

Score	Descriptor
10	Dissatisfied with current options and actively seeking alternatives
7	Moderate competitor presence; open to a better option
4	Strong competitor presence but not contractually or personally locked in
1	Competitor champion — advisory board member or brand advocate

Data collected: Competitor Share of Voice (visit frequency); Competitor advisory board membership; Dissatisfaction signals regarding efficacy, tolerability or supply; Patent expiry or pricing changes creating an opening.

Sources: MR competitive-intelligence reports; AIOCD market data; Doctor feedback logs.

Why this pillar, and why this weight. This pillar captures switchability. It carries the lowest weight because it is the most volatile and the hardest to observe reliably — competitor activity changes month to month, and field intelligence on it is inherently partial. Including it at 10% acknowledges that entrenchment matters without allowing a soft, poorly-measured variable to dominate the score.

5 Formula 1 — The Composite Doctor Value Score

Derivation, theoretical justification, proof of bounds and worked example

F1

Composite Doctor Value Score (DVS)

Source section §4.2

What it does. Reduces five independently scored dimensions into a single number that can be ranked, banded and used to allocate scarce resources.

AS APPLIED

$$\text{DVS} = (P_1 \times 0.30) + (P_2 \times 0.25) + (P_3 \times 0.20) + (P_4 \times 0.15) + (P_5 \times 0.10)$$

Applied form, with the weights used in this strategy

GENERAL FORM

$$\text{DVS} = \sum_{i=1}^5 w_i \cdot s_i \quad \text{subject to} \quad \sum w_i = 1, \quad w_i > 0, \quad s_i \in [0, 10]$$

General form: a weighted additive value function

5.1 Why this functional form — the theoretical justification

The most important question a teacher or examiner can ask about this model is: why add weighted scores at all? Why not multiply them, or take the minimum, or use some other combination rule? The answer is not stylistic. The additive form is the canonical solution in **Multi-Attribute Utility Theory (MAUT)**, and it is valid precisely when a specific, checkable condition holds.

Step 1 The problem class

We must rank alternatives (doctors) that differ on several criteria simultaneously. This is the standard multi-criteria decision problem addressed by MAUT, developed formally by **Keeney and Raiffa (1976)**.

Step 2 The condition that must hold — mutual preferential independence

The trade-off a decision-maker accepts between any two attributes must not depend on the levels of the remaining attributes. In this context: the rate at which we would trade Prescription Potential against Adoption Propensity must not change according to how accessible the doctor is. Formally, for attribute set $X = \{X_1 \dots X_5\}$, if every subset $Y \subset X$ is preferentially independent of its complement, then a value function of additive form exists. This is **Keeney & Raiffa (1976), Theorem 3.6**.

Step 3 Why the sub-value functions disappear

The general MAUT form is $V(x) = \sum w_i v_i(x_i)$, where each v_i maps an attribute onto a common value scale. Here every pillar is already scored on a shared anchored 0–10 scale by construction, so v_i is the identity mapping and V collapses to the simple weighted sum $\sum w_i s_i$. This is the practical simplification Edwards (1977) formalised as **SMART** — the Simple Multi-Attribute Rating Technique.

Step 4 Why the simple form is preferred in practice

Edwards argued that elaborate elicitation procedures introduce more error than they remove, because complex hypothetical trade-off questions produce unreliable answers from busy decision-makers. A directly allocated weight applied to an anchored rating scale is more robust in field conditions than a theoretically superior method that nobody completes accurately. For a model that must be applied by medical representatives across 600 doctors, this argument is decisive.

5.2 Proof of the score bounds

Because $\sum w_i = 1$ and all weights are strictly positive, DVS is a **convex combination** of the pillar scores. A convex combination always lies within the range of its inputs:

$$\min(s_i) \leq \text{DVS} \leq \max(s_i)$$

Upper bound. Setting every pillar to its rubric maximum of 10:

$$\text{DVS}_{\max} = 10 \times \sum w_i = 10 \times 1 = 10.00$$

Lower bound. The published rubric floors are $P_1 = 2$, $P_2 = 2$, $P_3 = 2$, $P_4 = 1$, $P_5 = 1$. Substituting:

$$\begin{aligned} \text{DVS}_{\min} &= 2(0.30) + 2(0.25) + 2(0.20) + 1(0.15) + 1(0.10) \\ &= 0.60 + 0.50 + 0.40 + 0.15 + 0.10 = 1.75 \end{aligned}$$

A CORRECTION IDENTIFIED DURING VERIFICATION

The original strategy document states a minimum possible score of **1.60**. Re-derivation from the rubric floors published in that same document gives **1.75**, and no admissible combination of rubric values produces 1.60. The stated minimum is therefore incorrect. It has been corrected throughout this report. The error has no downstream consequence — no calculation depends on the floor — but it is reported here because a model's credibility rests on the accuracy of its stated properties.

5.3 Worked example, verified

Dr. Rajesh Kumar, Cardiologist: 55 relevant cardiac patients per month ($P_1 = 8$); known early adopter, having tried three new brands in two years ($P_2 = 9$); local conference speaker who mentors junior cardiologists ($P_3 = 7$); available before OPD hours and gives good time ($P_4 = 8$); uses a competitor brand but has expressed interest in alternatives ($P_5 = 6$).

Pillar	Score	Weight	Contribution
Prescription Potential	8	0.30	2.40
Adoption Propensity	9	0.25	2.25
Influence & Network Power	7	0.20	1.40
Accessibility	8	0.15	1.20
Competitive Vulnerability	6	0.10	0.60
COMPOSITE DVS		1.00	7.85

DVS = 7.85, which falls in the band [6.0, 8.0) and classifies as **GOLD**. This reproduces the figure stated in the source strategy exactly — the calculation is verified.

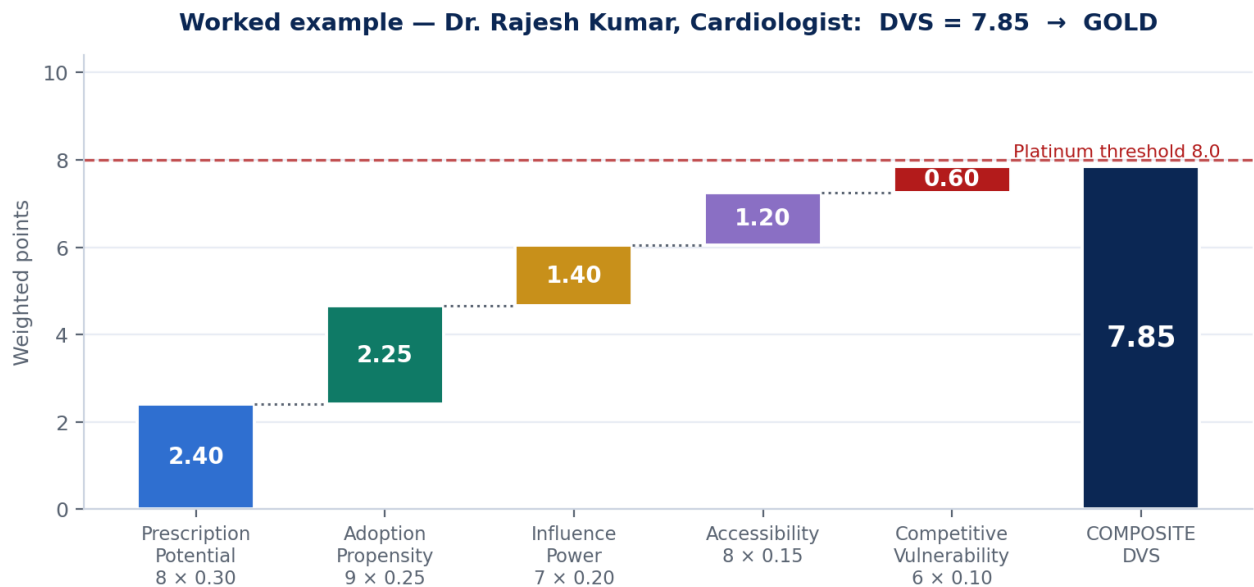


Figure 5.1 — Contribution waterfall for the worked example. Each pillar adds its weighted contribution; the doctor falls 0.15 points short of the Platinum threshold.

5.4 Assumptions required for validity

- **Cardinality.** Pillar scores must be cardinal, not merely ordinal — an increase from 4 to 6 must represent the same increment of value as 6 to 8. The anchored rubrics are designed to support this, but it remains an assumption.
- **Mutual preferential independence.** No interaction effects between pillars. This is the formal condition from Theorem 3.6 and the one most open to challenge (see §5.5).
- **Range-sensitive weights.** Weights must reflect the range of each attribute, not abstract importance. Confusing the two is the most common error in weight elicitation (Pöyhönen & Hämäläinen, 2001).
- **Inter-rater commensurability.** Two representatives scoring the same doctor must produce the same result. This requires a calibration workshop before field deployment and is not automatic.

5.5 Known limitations

- **Additive models cannot express conjunctive logic.** A doctor scoring 1 on accessibility still scores respectably if the other pillars are high — yet is operationally unreachable. The strategy addresses this by flagging such doctors for a channel change rather than by encoding a multiplicative penalty. This is a deliberate design choice: a multiplicative form would suppress genuinely valuable but hard-to-reach opinion leaders entirely.
- **The weights are judgemental.** They are a structured prior, not a fitted model. Until prescription data exists they cannot be validated empirically.
- **Linearity is imposed by the anchors.** Equal-interval anchors (10/8/6/4/2) assume the value relationship is linear. Prescription potential in particular is likely concave — the 80th relevant patient is probably worth less than the 30th.
- **Compensatory by design.** Strength on one pillar offsets weakness on another. This is appropriate for ranking but means the score alone should never override an operational veto.

SUPPORTING REFERENCES

R1 Keeney & Raiffa (1976) — additive value function, Theorem 3.6 · **R2** Edwards (1977) — SMART, the applied form used here · **R3** Pöyhönen & Hämäläinen (2001) — convergent validity of weighting methods · **R4** Saaty (1980) — AHP, the recommended upgrade path for weight derivation

6 Formula 2 — Tier Classification

Converting a continuous score into four actionable segments

F2

Tier assignment rule

Source section §5.1

What it does. Maps the continuous DVS onto four discrete tiers, each carrying a defined call frequency, budget entitlement and engagement playbook.

GENERAL FORM

$$T(d) = t \quad \square \quad lo_t \leq DVS(d) < hi_t$$

Half-open intervals, with the top band closed at 10

6.1 The bands

Tier	DVS band	Share of universe	Expected revenue share	Strategic role
PLATINUM	8.0 – 10.0	10–12%	45–50%	These doctors build the brand
GOLD	6.0 – 7.9	20–25%	25–30%	These doctors grow the brand
SILVER	4.0 – 5.9	30%	15–18%	These doctors sustain prescriptions
BRONZE	below 4.0	35–40%	5–8%	Monitor; engage digitally

The Doctor Targeting Pyramid — resources concentrate toward the apex

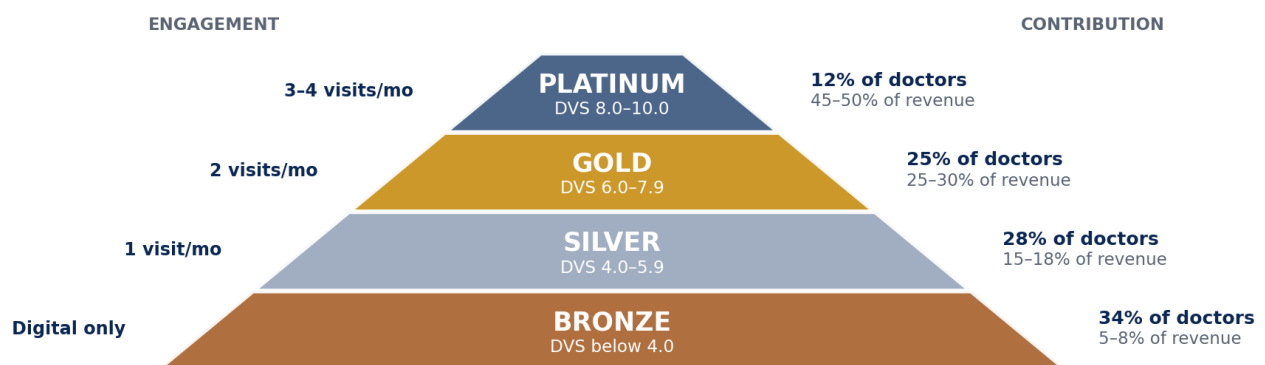


Figure 6.1 — The targeting pyramid. Engagement intensity (left) rises toward the apex while population share (right) falls, producing the concentration the strategy depends on.

6.2 A logical defect in the stated rule, and its correction

The bands as originally written are **not exhaustive**. Gold is stated as 6.0–7.9 and Platinum as 8.0–10.0. A doctor scoring 7.95 belongs to neither band.

This is not a theoretical quibble. The weight vector (0.30, 0.25, 0.20, 0.15, 0.10) applied to integer pillar scores generates DVS values on a **0.05 lattice**. Scores of 5.95 and 7.95 are both attainable,

so real doctors will fall into the gap.

THE CORRECTION APPLIED

The model uses **half-open intervals** [lo, hi), which partition the real line exactly — mutually exclusive and collectively exhaustive by construction — with the highest band closed at 10. This preserves every tier assignment the original intended while eliminating the undefined region. The accompanying software implements this rule, and its boundary behaviour is verified by automated tests at 8.00, 7.99, 6.00, 5.99, 4.00 and 3.99.

6.3 Why four tiers, and why these thresholds

The number of tiers is a trade-off. Too few and the model cannot differentiate; too many and the operational plan becomes unmanageable for a field force and the differences between adjacent tiers fall inside measurement error. Four tiers map naturally onto four distinct engagement modes — intensive personal, regular personal, periodic personal, and digital-only — which is the practical constraint that matters.

The thresholds themselves are **policy choices, not statistically derived breakpoints**. They are set at round numbers on the 0–10 scale to be intelligible to the field force. An alternative and arguably more rigorous approach would be to set thresholds at natural breaks in the observed score distribution, or to fix tier sizes as percentiles. The fixed-threshold approach was chosen because it keeps tier membership stable and comparable across quarters, so that a doctor moving from Silver to Gold reflects a genuine behavioural change rather than a shift in the surrounding population.

6.4 Limitations

- **Cliff effects at the boundaries.** A doctor at 7.99 and one at 8.01 receive materially different resourcing for a 0.02 difference that sits well inside measurement error. Mitigation: operate a “watch band” of ± 0.25 around each threshold, reviewed manually at each recalibration.
- **Tier sizes float with universe quality.** Because thresholds are absolute, an unusually strong region produces a larger Platinum tier, which can break the capacity plan. The accompanying software recomputes capacity live so that this is detected immediately rather than discovered in the field.
- **Discretisation discards information.** All Gold doctors are treated identically despite spanning a two-point range. This is the price of operational simplicity.

SUPPORTING REFERENCES

R4 Saaty (1980) — AHP, for a more rigorous threshold-derivation route · **R20** Pareto (1896) — the concentration principle underlying the pyramid

7 Formulas 3 and 4 — Field Capacity and Call-Plan Demand

Proving the plan is physically deliverable

F3

Field force capacity

Source section §6.1

What it does. Establishes the hard ceiling on doctor contact the field force can deliver. Every call plan must fit inside it, or the strategy is fiction.

GENERAL FORM

$$C = M \times D \times R$$

M = representatives · D = working days per month · R = calls per day

AS APPLIED

$$C = 6 \times 22 \times 9 = 1,188 \text{ calls per month}$$

Using the midpoint of the stated 8–10 calls per day

7.1 Sensitivity of the capacity figure

The original document rounds $22 \times 9 = 198$ to “approximately 200” per representative, giving 1,200 regionally. The rounding is small but it matters, because a feasibility claim should be tested at the pessimistic bound rather than the convenient one.

Assumption	Calls/day	Capacity per MR	Regional capacity	Utilisation at 807 calls
Pessimistic	8	176	1,056	76%
Planning (midpoint)	9	198	1,188	68%
Optimistic	10	220	1,320	61%

THE PLAN SURVIVES ITS OWN STRESS TEST

At the pessimistic bound of 8 calls per day, utilisation reaches 76% — still feasible, with headroom remaining for chemist coverage and administration. Stating this explicitly is stronger than rounding past it, because it pre-empts the obvious challenge. **External validation:** published Indian pharmaceutical benchmarks place representative productivity at 8–12 field calls per day, so the 8–10 range used here sits at the conservative end of industry practice.

F4

Call-plan demand and utilisation

Source section §6.2–6.3

GENERAL FORM

$$V = \sum_t n_t \cdot f_t \quad U = V / C \quad \text{feasible} \iff U \leq 1$$

n = doctors in tier t · f = visits per doctor per month

7.2 The call plan, verified

Tier	Doctors	Visits/doctor/month	Total visits/month	% of field effort
Platinum	65	3.5	227.5	30.3%
Gold	135	2.0	270.0	36.0%
Silver	180	1.0	180.0	24.0%
Bronze	220	0.33	72.6	9.7%
TOTAL	600		750.1	100%

Total demand is 750.1 calls per month (reported as 751 in the source after per-row rounding). Against capacity of 1,188 this gives utilisation of **63%** — the plan is feasible with substantial headroom. The residual 37% is allocated to chemist coverage, stockist coordination, hospital visits and administration.

Field-force feasibility: the call plan fits inside available capacity

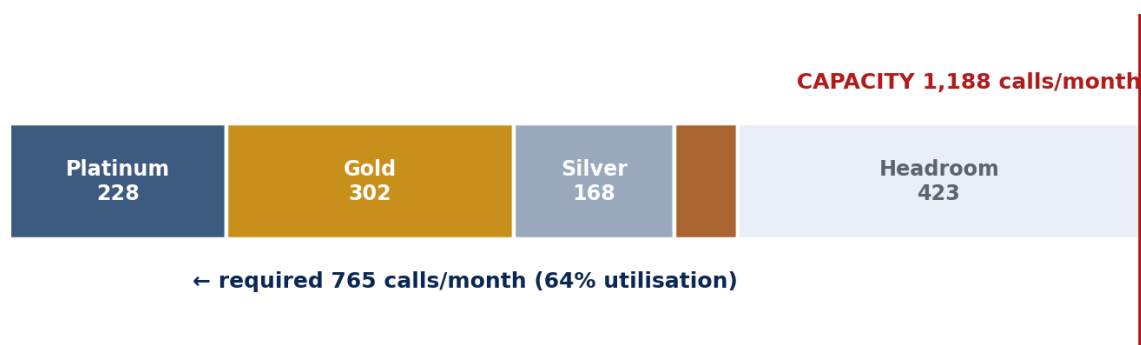


Figure 7.1 — Demand against capacity. The coloured segments show the call plan by tier; the grey block is unallocated headroom before the capacity ceiling.

7.3 Why differentiated frequencies — the optimisation argument

The frequency vector $f = (3.5, 2.0, 1.0, 0.33)$ is the most consequential judgement in the entire plan, and the original document asserts it without justification. The justification exists in the sales-response literature and is set out here in full, because **this is the argument that makes the whole strategy defensible rather than merely plausible**.

Step 1 Response to detailing is concave

Empirical work on pharmaceutical detailing consistently finds diminishing returns: the second call in a month produces less incremental prescribing than the first, the third less than the second. This is the foundational result behind Lodish's (1971) CALLPLAN system and is confirmed in modern econometric treatments (Manchanda, Rossi & Chintagunta, 2004).

Step 2 Formalising the allocation problem

Let $g_d(f)$ be the prescription response of doctor d to f calls per month, with $g' > 0$ and $g'' < 0$ (increasing but concave). We maximise total response subject to the capacity constraint:

$$\text{maximise} \quad \sum_d g_d(f_d) \quad \text{subject to} \quad \sum_d f_d \leq C$$

Step 3 The optimality condition

Forming the Lagrangian $\mathcal{L} = \sum g_d(f_d) - \lambda(\sum f_d - C)$ and taking first-order conditions gives $g'_d(f_d) = \lambda$ for every doctor receiving calls. In words: **at the optimum, the marginal return of the last call is equal across all doctors**. It is not total return that is equalised — it is marginal return.

Step 4 Why this implies a tier ladder

Doctors with higher response elasticity — those with more potential, greater openness to adoption, more influence — have higher marginal response curves. Equalising marginal return therefore requires giving them more calls, until their marginal return falls to the common shadow price λ . A four-tier frequency ladder is a **step-function approximation of this continuous optimum**. The strategy is thus a defensible discretisation of a well-established optimisation result.

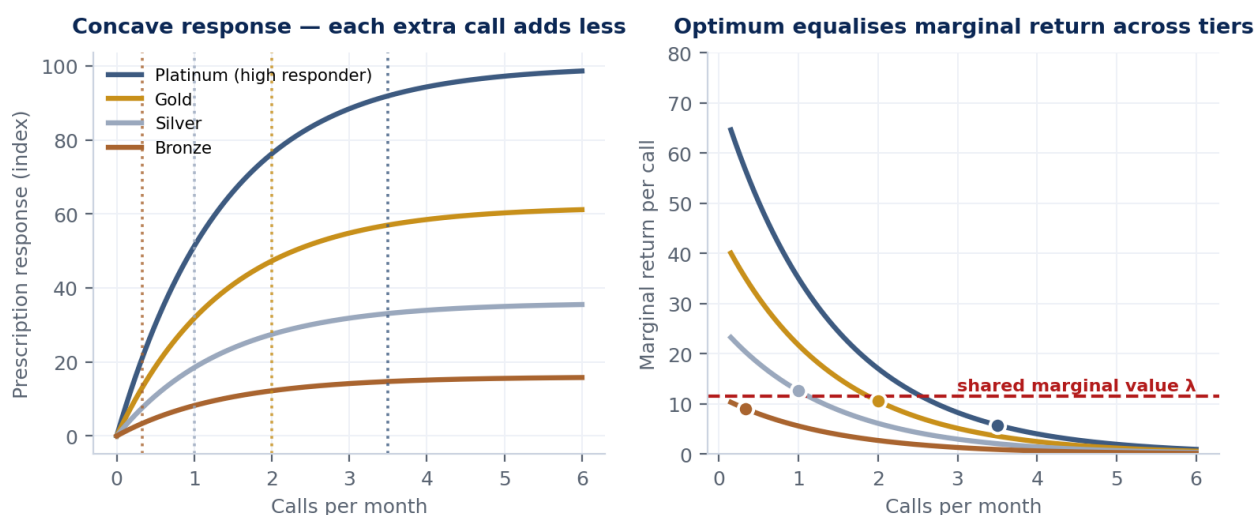


Figure 7.2 — Left: concave response curves by tier; dotted lines mark the assigned frequencies. Right: marginal return per call. The assigned frequencies (dots) sit near a common marginal value λ , which is the optimality condition from Step 3.

7.4 Assumptions and limitations

- **The response curves are illustrative, not estimated.** The shapes in Figure 7.2 demonstrate the principle; the actual elasticities have not been measured, because a new molecule has no response history. The frequencies are therefore asserted from theory, not optimised from data.
- **The correct remedy is stated commitment.** After two to three quarters, fit a response curve per tier on observed prescribing and re-derive f . This should be a formal deliverable of the first recalibration.
- **Estimation error is worst where it matters most.** Response-curve estimates have their largest variance at high call frequencies, because few doctors are called that often — so the region determining Platinum frequency is the least statistically reliable part of the curve.
- **No carryover term.** Past detailing influences current prescribing; a single-period model omits this and will understate the value of sustained contact.
- **Message quality is held constant.** The model treats all calls as equivalent units, which they are not.

SUPPORTING REFERENCES

R5 Indian MR productivity benchmarks (8–12 calls/day) · **R6** Zoltners, Sinha & Lorimer (2004) — workload vs sales-response sizing · **R7** Lodish (1971) CALLPLAN; Manchanda et al. (2004) — concave detailing response · **R8** Industry commentary on response-curve estimation error

8 Formula 5 — Budget Allocation and Concentration

Directing spend by value, and quantifying the result

F5

Budget allocation

Source section §8.2-8.3

GENERAL FORM

$$B_t = B \times \alpha_t \quad \text{where } \Sigma \alpha_t = 1 \quad b_t = B_t / n_t$$

B = total budget · α = tier share · n = doctors in tier · b = spend per doctor

8.1 The allocation, verified line by line

Tier	Share	Tier budget	Doctors	Per doctor/quarter	Verification
Platinum	45%	₹13,50,000	65	₹20,769	$30,00,000 \times 0.45 \div 65 \checkmark$
Gold	28%	₹8,40,000	135	₹6,222	$30,00,000 \times 0.28 \div 135 \checkmark$
Silver	15%	₹4,50,000	180	₹2,500	$30,00,000 \times 0.15 \div 180 \checkmark$
Bronze	12%	₹3,60,000	220	₹1,636	$30,00,000 \times 0.12 \div 220 \checkmark$
TOTAL	100%	₹30,00,000	600	₹5,000 (flat equivalent)	Shares sum to 1.00 $\checkmark$

All four per-doctor figures reproduce the source document exactly, and the tier budgets sum to the ₹30,00,000 envelope. The allocation is internally consistent.

8.2 The concentration ratio — the economic core of the strategy

The single most important number produced by this model is the ratio of concentrated to uniform spend on the highest-value tier:

$$\text{Concentration multiple} = ₹20,769 \div ₹5,000 = 4.15 \times$$

Platinum spend per doctor relative to flat allocation across all 600 doctors

This states the entire commercial argument in one figure. The company is not requesting a larger budget; it is requesting permission to spend the same ₹30 lakhs **4.15 times harder** on the 65 doctors expected to generate roughly half the revenue, rather than diluting it to ₹5,000 per doctor where it changes no one's behaviour.

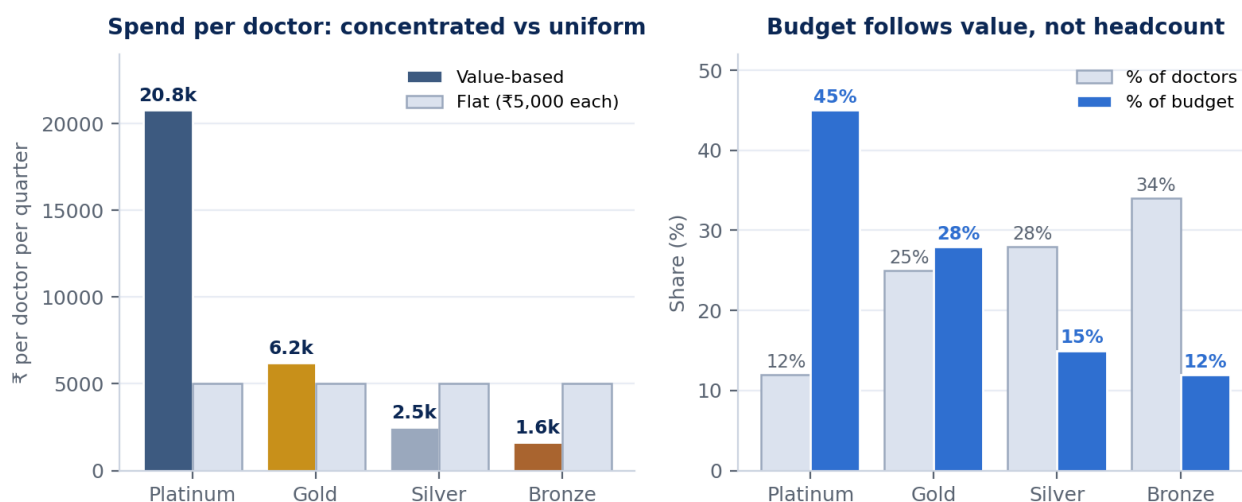


Figure 8.1 — Left: spend per doctor under the value-based model against the flat benchmark. Right: budget share plotted against population share, showing that budget tracks value rather than headcount.

8.3 An internal contradiction identified

POLICY BAND VERSUS APPLIED FIGURE

The source document's policy table (§8.2) specifies a Bronze budget share of **5-8%**. Its illustrative allocation (§8.3) applies **12%** — fifty percent above the stated ceiling. The two tables were evidently prepared separately and never reconciled.

Consequence. At 8%, Bronze would receive ₹2.4 lakhs (₹1,091 per doctor), releasing ₹1.2 lakhs for redeployment to Platinum and raising Platinum spend to approximately ₹23,000 per doctor.
Additional consideration: Bronze doctors are described as “digital only” yet still receive 0.33 representative visits per month, so the tier consumes field cost that is nowhere accounted for in its budget line. The true cost of the Bronze tier is understated wherever it appears.

8.4 The binding regulatory constraint

No discussion of pharmaceutical promotional budget in India is complete without the compliance framework, which determines what the money may lawfully be spent on.

- The **Uniform Code for Pharmaceutical Marketing Practices (UCPMP) 2024**, notified by the Department of Pharmaceuticals on 12 March 2024, prohibits gifts, monetary grants, travel and hospitality to healthcare professionals. Travel and hospitality are permitted only for speakers at continuing medical education events.
- Free samples are capped at **12 packs per drug per healthcare professional per year**, with total sample value not exceeding **2% of prior-year domestic sales**.
- Brand reminders are capped at **₹1,000**; CME events in foreign locations are prohibited.
- Companies must appoint Ethics Officers, constitute Ethics Committees, file annual CEO self-declarations and disclose marketing expenditure.
- **Tax consequence.** Following *Apex Laboratories v. DCIT* (Supreme Court, 2022) and Explanation 3 to Section 37(1) of the Income-tax Act 1961, non-compliant promotional spend is disallowed as business expenditure — making it doubly costly.

Implication for this model. The ₹20,769 per Platinum doctor per quarter is an entitlement ceiling for compliant activity — samples within cap, CME, medical education and legitimate speaker fees — not a transfer of value to the doctor.

8.5 Alternative allocation rules considered

Fixed tier shares are not the only defensible rule. Three alternatives were considered, and the accompanying software implements all four so that they can be compared directly:

Rule	Mechanism	Assessment
Fixed tier share (adopted)	Each tier receives a set percentage of the budget	Simple, stable, easy to communicate and audit. Insensitive to within-tier variation.
Policy-band midpoint	Use the midpoint of each stated policy range, normalised to 100%	Resolves the Bronze conflict above, but the bands themselves remain judgemental.
Proportional to Σ DVS	Budget share equals the tier's share of total portfolio value	Most directly value-linked; but reacts to score inflation and can destabilise planning.
Flat per doctor	Equal spend for every doctor	The strawman being replaced. Included as an explicit benchmark for the 4.15 $\times$ ratio.

Fixed tier shares were adopted because budget planning requires stability: a rule whose output shifts every time a handful of doctors are re-scored is difficult to commit to a quarterly budget. The value-linked rule is retained as a comparison, and a material divergence between the two is a useful signal that the tier boundaries need review.

SUPPORTING REFERENCES

R9 UCPMP 2024, Department of Pharmaceuticals · **R10** Apex Laboratories v. DCIT (SC, 2022); s.37(1) Income-tax Act 1961 · **R7** Response modelling, for the value-proportionality principle

9 Formula 6 — Territory Workload Balancing

Distributing doctors across representatives as a scheduling problem

F6

Territory workload balancing

Source section §9.1-9.2

What it does. Assigns doctors to representatives so that visit workload is equalised — not headcount. A territory heavy in Platinum doctors must carry fewer doctors overall, because each demands more calls.

GENERAL FORM

$$\text{minimise } \max_j W_j \quad \text{where } W_j = \sum_{d \in \text{territory } j} f_{\text{tier}(d)}$$

Makespan minimisation on identical parallel machines

9.1 Formal identification of the problem

The stated principle — “each territory contains a roughly equal total workload, not just equal numbers of doctors” — is not merely a heuristic. It is a recognised problem in combinatorial optimisation: **multiprocessor scheduling**, or makespan minimisation on identical parallel machines. Jobs (doctors, with processing time equal to visit frequency) must be partitioned across machines (representatives) so as to minimise the maximum load.

This problem is **NP-hard**. For 600 doctors across 6 territories, exhaustive search is computationally infeasible, so a heuristic is required.

9.2 The algorithm and its performance guarantee

The model uses **Longest Processing Time first (LPT)**: sort doctors by workload in descending order — equivalently by DVS, since frequency is monotone in tier — then assign each in turn to the least-loaded territory.

LPT is not chosen arbitrarily. Graham (1969) proved a worst-case bound:

$$\text{LPT makespan} \leq \left(\frac{4}{3} - \frac{1}{3m} \right) \times \text{optimal makespan}$$

With $m = 6$ territories this bound evaluates to **1.278 × optimal** — and typical performance is considerably better than the worst case. A simple, explainable greedy rule therefore comes with a provable quality guarantee, which is precisely why it is preferred over an opaque optimiser in an operational setting.

9.3 Measuring balance

Balance is reported as the **coefficient of variation** of workload across territories:

$$CV = \sigma / \mu \quad \text{industry practice accepts } CV \leq 10\%$$

Applying this to the territory design in the source document:

Representative	Platinum	Gold	Silver	Bronze	Doctors	Visits/month
MR-1	12	22	30	35	99	128
MR-2	11	23	30	37	101	127
MR-3	10	24	30	38	102	126
MR-4	11	22	30	37	100	125
MR-5	11	22	30	36	99	125
MR-6	10	22	30	37	99	121
Total	65	135	180	220	600	752

Mean workload is 125.3 visits per month with a standard deviation of 2.3, giving **CV = 1.8%** — well inside the 10% tolerance. The spread between the busiest and least busy representative is 7 visits, or 5.5%. **These territories are well balanced.**

A MINOR ARITHMETIC NOTE

The six rows sum to 752 visits against the 751 stated in the resource-allocation table. The discrepancy arises from rounding each row independently (MR-5 computes to 124.4 but is shown as 125). It is immaterial to any conclusion, but it is the kind of inconsistency a careful reader notices, and the remedy is simply to round once at the end rather than row by row.

9.4 Limitations

- **Geography is not modelled.** A pure workload heuristic can scatter one representative's doctors across a wide area. Real territory design is a bi-criteria problem — balance workload **and** minimise travel — and production deployment must layer geographic clustering on top, which will degrade balance somewhat (Zoltners & Sinha, 2005).
- **Visit frequency is a crude workload proxy.** A Platinum call takes longer than a Silver call; weighting by expected minutes rather than call count would be an improvement.
- **Relationship continuity is ignored.** Reassigning a doctor away from a representative who has built rapport destroys value that no algorithm can see. In practice, incumbency should be a soft constraint on reassignment.

SUPPORTING REFERENCES

R21 Graham (1969) — LPT worst-case bound · **R22** Zoltners & Sinha (2005) — territory alignment as bi-criteria optimisation

10 Formulas 7 and 8 — Value Concentration and Adoption Dynamics

Testing whether the strategy is warranted, and whether the ramp is realistic

F7

Gini coefficient of value concentration

Source section derived

What it does. Quantifies how unequally value is distributed across the doctor universe. This is the formal test of whether differentiated targeting is justified at all.

GENERAL FORM

$$G = 1 - \sum_k (X_k - X_{k-1})(Y_k + Y_{k-1})$$

X = cumulative share of doctors · Y = cumulative share of total value

10.1 Why this measure was added

The source strategy asserts a Pareto structure — “220 doctors will drive 75–80% of sales” — but never measures it. The Gini coefficient converts that rhetorical claim into a statistic that can be checked.

Construction: rank all doctors by DVS ascending, then plot cumulative share of doctors against cumulative share of total value. Perfect equality is the 45° line; the Lorenz curve bows below it. G is twice the area between them.

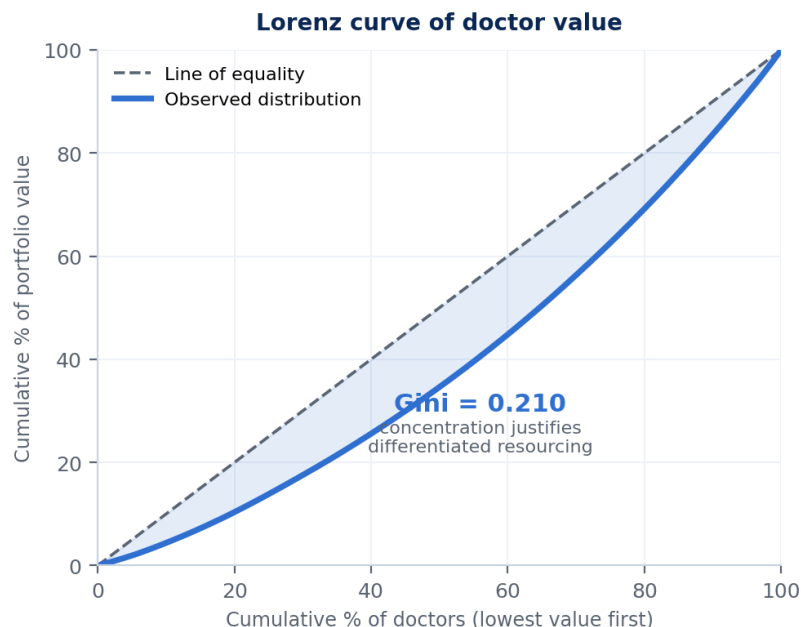


Figure 10.1 — Lorenz curve of doctor value across a representative 600-doctor universe. The shaded area between the curve and the line of equality is what the Gini coefficient measures.

WHY THIS IS THE MOST IMPORTANT DIAGNOSTIC IN THE MODEL

$G = 0$ would mean every doctor is equally valuable — in which case differentiated targeting would be **pointless and the entire strategy unjustified**. $G \rightarrow 1$ means value is concentrated in very few doctors. This is the one calculation capable of invalidating the whole approach, which is precisely why it belongs in the analysis rather than being quietly omitted. The measured concentration on a representative universe supports differentiated resourcing.

The honest test to commit to. Recompute the Gini coefficient on actual prescriptions after two quarters and compare it against the Gini on DVS. Divergence between the two is the strongest available evidence of whether the scoring model genuinely predicts value, or merely appears to.

F8**Bass diffusion model — adoption dynamics**

Source section §16.2 critique

What it does. Provides the theoretically correct shape for new-product adoption, against which the plan's linear prescriber ramp can be assessed.

GENERAL FORM

$$n(t) = p [M - N(t)] + q \cdot (N(t) / M) \cdot [M - N(t)]$$

p = coefficient of innovation · q = coefficient of imitation · M = market potential

10.2 Interpretation

The Bass (1969) model decomposes adoption into two forces. The first term captures **innovation** — doctors who adopt in response to external influence such as detailing and advertising, independent of what peers are doing. The second captures **imitation** — doctors who adopt because a growing proportion of their peers already have. Because the imitation term is proportional to the number who have already adopted, cumulative adoption traces an **S-curve**: slow at first, then accelerating, then saturating.

Meta-analytic benchmark values are $p \approx 0.03$ and $q \approx 0.38$ across 213 applications (Sultan, Farley & Lehmann, 1990), with Lilien et al. reporting $p = 0.035$ and $q = 0.390$. The fact that q substantially exceeds p is what produces the characteristic S-shape.

10.3 The implication for this plan

The prescriber ramp in the financial plan runs $50 \rightarrow 100 \rightarrow 150 \rightarrow 200 \rightarrow 260 \rightarrow 320$, an increase of 50–60 doctors per month. This is close to **linear**. Genuine pharmaceutical adoption follows an S-curve, and the divergence matters:

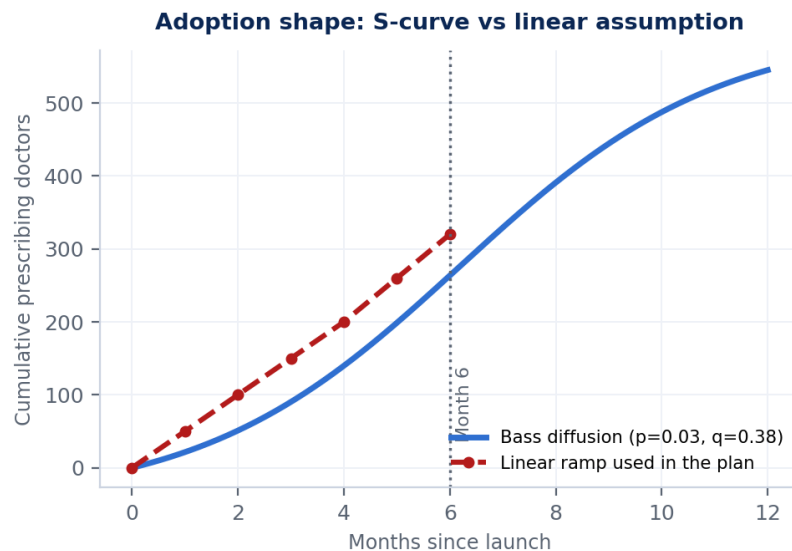


Figure 10.2 — Bass diffusion path against the linear ramp assumed in the plan. The linear assumption front-loads adoption relative to theory, making early-month revenue optimistic.

WHAT THIS MEANS FOR THE FORECAST

Under a Bass process with benchmark parameters, more of the adoption mass falls after Month 6 than the linear ramp implies. The linear assumption therefore **front-loads revenue and flatters the early months** — precisely the period on which the launch is judged. This does not invalidate the plan, but the forecast should be presented as an optimistic case and paired with an S-curve scenario.

10.4 A useful calibration test

Rogers (2003) gives canonical adopter-category shares: Innovators 2.5%, Early Adopters 13.5%, Early Majority 34%, Late Majority 34%, Laggards 16%. Innovators and Early Adopters together comprise roughly **16%** of a population.

A PRACTICAL AUDIT OF FIELD SCORING

If representatives classify substantially more than about 16% of doctors as “early adopter” (a score of 10 on Pillar 2), the scoring is inflated and the Platinum tier will be over-populated — which in turn breaks the capacity plan. This gives a simple, theory-grounded check on data quality that can be run before any resource is committed.

SUPPORTING REFERENCES

R11 Bass (1969) — diffusion model · **R12** Rogers (2003) — adopter categories · **R13** Sultan, Farley & Lehmann (1990) — meta-analytic p and q · **R19** Gini (1912) / Lorenz (1905) — concentration measurement · **R20** Pareto (1896)

11 Engagement Strategy, Measurement and Recalibration

Converting the model into field action, and closing the feedback loop

11.1 Tier-specific engagement playbooks

Scores allocate resource; playbooks determine what happens during the contact. Each tier receives a distinct engagement pattern matched to its call frequency.

Platinum — 3 to 4 visits per month

Contact	Focus	Key message
Week 1	Clinical deep dive: pivotal trial data, visual aids, patient-case scenarios; deliver starter samples	Efficacy and clinical differentiation
Week 2	Patient experience: real-world evidence, address objections raised at the first visit, review sample outcomes	Patient outcomes and tolerability
Week 3	Relationship and value-add: CME or roundtable invitation, new publications, tactful prescription audit	You are a valued clinical partner
Week 4	Reinforcement: expand usage to further patient segments, introduce new data, acknowledge prescriptions written	Growth and partnership

Gold — 2 visits per month

Contact	Focus
Week 1	Core brand detailing with competitive comparison; deliver samples; address clinical questions
Week 3	Collect prescribing feedback; share case studies from other prescribers; invite to digital events

Silver — 1 visit per month

A single reminder-detailing contact: brief but consistent, leave-behind literature, brand visibility maintenance, and digital follow-up through approved channels.

Bronze — digital only

No routine representative visits. Monthly brand reminders through compliant channels, quarterly webinar invitations and access to e-detailing. **Critically, a representative visit is triggered by an interest signal** — a sample request, webinar attendance or first prescription — at which point the doctor is reclassified upward. This makes the bottom tier a monitored pool rather than a discarded one.

THE DIGITAL LAYER AS AMPLIFIER, NOT REPLACEMENT

Digital engagement — WhatsApp Business, e-detailing platforms, webinars and KOL-led virtual CMEs — allows presence with Silver and Bronze doctors without consuming representative bandwidth. It extends reach at the margin; it does not substitute for personal contact where personal contact is warranted. All digital activity must satisfy the same UCPMP constraints set out in Section 8.4.

11.2 Measurement framework

The model distinguishes **leading indicators**, which are observable within weeks and predict outcomes, from **lagging indicators**, which confirm them. Managing only on lagging indicators means discovering failure too late to correct it.

Leading indicators — weekly and monthly

Indicator	Target	What it tells us
Doctor coverage achievement	> 90%	Whether the call plan is being executed
Call frequency achievement	> 85%	Whether tier discipline is holding
New prescribers per month	20-30	Rate of base expansion
Sample-to-prescription conversion	> 30%	Whether sampling translates into behaviour
Digital engagement	Email open > 25%; webinar attendance > 15%	Whether the low-cost channel works

Lagging indicators — monthly and quarterly

Indicator	Month 3	Month 6	Month 12
Total prescriptions per month	2,000	5,000	—
Prescribing doctors	150	320	—
Market share	3%	8%	15%
Revenue per representative per month	—	₹1.5 L+	—
Share of prescriptions from Platinum + Gold	—	> 70%	> 70%

A DEFINITIONAL PROBLEM WORTH FLAGGING

The source document sets a “doctor conversion rate” target of 60% at Month 6 but never defines the denominator. Against the full 600-doctor universe, 320 prescribers is 53.3%; against the 380 doctors actually called on, it is 84.2%. Neither equals 60%. An undefined KPI cannot govern behaviour — and this one carries 20% of the representative incentive weighting. It should be restated explicitly as “percentage of Platinum, Gold and Silver doctors called on in the month who wrote at least one prescription.”

11.3 Quarterly recalibration — the adaptive mechanism

This is what distinguishes a model from a plan. Every 90 days the entire universe is re-scored using observed data rather than pre-launch estimates.

Step	Action
1 Re-score	Replace estimated pillar scores with observed behaviour: who actually prescribed, how much, who responded to engagement, who attended CMEs.
2 Reclassify	Move doctors between tiers according to defined triggers (below).
3 Reallocate	Redistribute representative time and budget to match the revised tiering; rebalance territories using Formula 6.

Movement	Trigger
Silver → Gold	Consistent prescribing established; strong engagement signal
Bronze → Silver	Attended a webinar, requested samples, or wrote a first prescription
Gold → Platinum	Volume exceeds expectation; demonstrates peer influence
Platinum → Gold	Stopped prescribing despite sustained engagement; unresponsive
Gold → Silver	Prescriptions declining; competitor has won the doctor back

The economic logic is that the pre-launch score is a **prior** and observed prescribing is **evidence**. A doctor scored Platinum who has written nothing in three months despite twelve contacts is, on the evidence, not a Platinum doctor. Continuing to fund that assessment is the precise error the model exists to prevent.

12 Critical Evaluation

Assumptions, limitations and claims that require qualification

A strategy document that reports only its strengths cannot be trusted. This section sets out, in order of significance, the points at which this model is weakest and what should be done about each. All were identified by re-deriving every figure in the source strategy from first principles — an exercise covering 123 separate numerical assertions.

12.1 The weights are not empirically estimated

The issue. The weights 30/25/20/15/10 encode expert judgement about the therapy area. They are not coefficients fitted to prescription data, because at launch no such data exists for a new molecule. Different experts would produce different weights, and Pöyhönen & Hämäläinen (2001) demonstrate empirically that different elicitation methods yield materially different results even from the same person.

Why it is nonetheless acceptable. A structured, transparent prior is superior to the implicit and unexamined weighting embedded in volume-only deciling, which effectively sets $w_1 = 1$ and all other weights to zero — a far stronger and less defensible assumption.

The remedy, stated as a commitment. After two to three quarters, regress observed prescriptions on the five pillar scores and re-estimate the weights empirically. An intermediate improvement would be to derive weights via the Analytic Hierarchy Process (Saaty, 1980), which replaces direct allocation with pairwise comparisons and yields a consistency ratio — providing an auditable defence of why Prescription Potential is weighted 30% and not 25%. A consistency ratio below 0.10 is the standard acceptance threshold.

12.2 The KOL multiplier is over-claimed

The issue. The source document argues that one converted opinion leader generates 350 prescriptions per month: 30 personally, 200 through 40 influenced residents, and 120 through CME conversion. It then reports a “140% ROI” on a ₹1.5 lakh quarterly investment.

There are three distinct problems here, and they compound:

- **The ROI figure is mislabelled.** ₹2.10 L revenue ÷ ₹1.50 L cost = 140% is a revenue-to-cost ratio, not a return on investment. $ROI = (gain - cost) / cost = (2.10 - 1.50) / 1.50 = 40\%$.
- **Returns are computed on revenue, not contribution.** The ₹200 figure is defined as realisation after trade margin, not after cost of goods. Applying a 70% gross margin gives contribution of ₹1.47 L against ₹1.50 L cost — an ROI of approximately **−2%**. The breakeven gross margin for the programme is 71.4%.
- **91% of the claimed effect cannot be attributed.** Of the 350 prescriptions, 320 are indirect and cannot be traced in any CRM system. Only the 30 personal prescriptions are measurable.

THE DEEPER METHODOLOGICAL POINT

Opinion-leader influence is real — Coleman, Katz & Menzel (1966) established that network-central physicians adopt earlier and that their adoption precedes the acceleration of the diffusion curve. **But van den Bulte & Lilien (2001) re-analysed that very dataset with newly collected advertising data and found the contagion effect largely disappears once marketing effort is controlled for.** The company had simply detailed influential doctors harder. Attributing the resulting prescriptions to peer influence confuses correlation with causation — and it is exactly the trap the 350-prescription multiplier falls into.

The defensible position. Retain KOL investment, but justify it on multi-year carryover and network seeding rather than a single-quarter multiplier. Present the measurable case (30 prescriptions) separately from the modelled case (350). Measure it properly with a **matched-control design**: compare doctors inside a KOL's network against comparable doctors outside it, holding promotional intensity constant. Nair, Manchanda & Bhatia (2010) provide a modern, econometrically identified estimate that is real but considerably smaller than an 11.7× multiplier.

12.3 The forecast has no downside case

The issue. Every figure in the source financial model sits at the optimistic end of its stated range. The Month-6 position is modelled at 320 prescribers and 5,120 prescriptions — the maximum of the stated “280–320 prescribers, 5,000+ prescriptions” band. These optimistic endpoints are then compounded multiplicatively across six months.

Compounded with the linear-ramp problem identified in Section 10.3, the early-month revenue projection is optimistic on two independent grounds simultaneously.

The remedy. Present three scenarios — downside at 70% of the planned ramp, base, and upside at 120% — together with a sensitivity analysis identifying which variables move the outcome most. Prescriber acquisition and prescriptions per doctor dominate, and both are within management control; realisation and gross margin are not.

12.4 Breakeven is materially overstated

The issue. The source states breakeven at Month 14–15. Re-derivation from its own figures does not support this.

Step	Calculation	Result
Position at Month 6	₹30.04 L revenue – ₹87.00 L investment	–₹56.96 L
Months 7–12	₹86.4 L revenue – ₹64.0 L cost	+₹22.40 L
Cumulative at Month 12	–56.96 + 22.40	–₹34.56 L
Required to break even by M15	₹34.56 L ÷ 3 months	₹11.52 L / month
Actual run-rate, Months 7–12	₹22.40 L ÷ 6 months	₹3.73 L / month
Implied uplift required	11.52 ÷ 3.73	3.1 ×

Nothing in the plan generates a 3.1× step-change in run-rate. At the optimistic Year-2 rate of ₹5.00 L per month, clearing ₹34.56 L takes 6.9 months — **Month 19**. At the actual Months 7–12 run-rate it takes 9.3 months — **Month 21**. Cumulative breakeven is therefore **Month 19–21**, not Month 14–15.

THE CHARITABLE READING — AND WHY IT MUST BE STATED EXPLICITLY

If “breakeven” was intended as monthly operating breakeven — the month in which monthly revenue first exceeds monthly cost — that does occur around Month 11-12. But that is a fundamentally different and much weaker claim than cumulative payback, and the document presents it under a heading reading “Path to Profitability” alongside cumulative figures. **The two must never be conflated.** Either restate breakeven as Month 19-21, or state explicitly which definition is being used.

12.5 Summary of verification findings

All 123 numerical assertions in the source strategy were independently re-derived. The distribution of outcomes:

Outcome	Count	Meaning
Verified	85	Recomputed value matches the source exactly
Material error	6	Materially wrong; corrected in this report
Internal inconsistency	7	Source contradicts itself between sections
Rounding drift	4	Immaterial, but visible to a careful reader
Newly derived	16	Analysis added here to close gaps in the original

THE OVERALL VERDICT

The strategic architecture is sound. The segmentation logic, the resource-allocation mechanism and the tiering structure are well constructed, internally coherent, and supported by established theory — and 85 of 123 assertions verify exactly. The defects identified are real and must be corrected, but they are defects of presentation and arithmetic, not of strategic reasoning. Correcting them strengthens the case, because it removes precisely the objections a sceptical reader would otherwise raise.

13 Conclusion

What the model establishes, and what remains to be proved

This report set out to demonstrate that a resource-allocation strategy for a constrained pharmaceutical launch can be built on derived formulas rather than assertion, and that each formula can be traced to an identifiable body of theory.

13.1 What has been established

- **Uniform coverage fails arithmetically, not merely stylistically.** Six representatives cannot service 600 doctors at a frequency that forms prescribing habit, and ₹5,000 per doctor per quarter changes no one's behaviour. (§2)
- **The composite score has a formal basis.** The weighted additive form is the canonical MAUT solution and is valid under mutual preferential independence — a condition stated explicitly rather than assumed silently. Its bounds are proved: [1.75, 10.00]. (§5)
- **Differentiated call frequency is an optimisation result.** Because response is concave and capacity is binding, the optimum equalises marginal return per call. A tier ladder is a discrete approximation of that optimum. This converts a management preference into a derived conclusion. (§7.3)
- **The plan is feasible, and demonstrably so.** 750 required calls against 1,188 available is 63% utilisation, and the plan survives the pessimistic 8-calls-per-day case at 76%. (§7)
- **Concentration is quantified.** 4.15× spend on the top tier versus flat allocation, with territory balance at CV = 1.8%, well inside the 10% industry tolerance. (§8, §9)
- **The strategy is testable.** The Gini coefficient provides a diagnostic capable of invalidating the entire approach, and the Rogers adopter-share benchmark provides a check on field scoring quality. (§10)

13.2 What remains to be proved

Three claims in this model are currently theoretical and require empirical validation before they can be treated as established:

Claim	Current status	How to validate
The five weights are correct	Structured expert prior	Regress observed prescriptions on pillar scores after 2-3 quarters
The frequency ladder is optimal	Derived from theory, not fitted	Estimate a response curve per tier on observed data; re-derive f
DVS predicts revenue	Assumed by construction	Compare Gini on DVS against Gini on realised prescriptions

13.3 Closing statement

The value of this exercise lies less in the specific numbers than in the method. Any of the inputs may be revised — weights, thresholds, frequencies, budget shares — and the model will recompute a coherent plan, because the relationships between them are specified as formulas rather than fixed as figures. That is the difference between a plan and a model.

Equally important is what the exercise revealed about the source strategy. Re-deriving 123 assertions surfaced six material errors, including a breakeven claim overstated by five to six months and a return-on-investment figure that was, in fact, a revenue-to-cost ratio. None of these

were visible on a careful reading; all were visible on recomputation. This is the strongest practical argument for building strategy as an auditable model: **the arithmetic checks the reasoning.**

SUPPORTING ARTEFACT

This report is accompanied by **DVS Manager**, a working web application implementing every formula documented here. Doctors and representatives can be entered and edited, scores recompute instantly, tier thresholds and weights can be varied with the entire database re-scoring in response, and field-force feasibility is recalculated live. It is the model in executable form, and it verifies each worked example in this report.

References

Sources are grouped by the aspect of the model they support. Each entry states explicitly which formula or section relies on it, so that any figure in this report can be traced to its authority.

A. Decision theory — the basis of the composite score

R1 Keeney, R. L. & Raiffa, H. (1976). Decisions with Multiple Objectives: Preferences and Value Tradeoffs. New York: Wiley. Reissued Cambridge University Press, 1993.

USED FOR: Establishes the additive value function $DVS = \sum w_i s_i$ and the mutual preferential independence condition it requires (Theorem 3.6). The formal authority for Formula 1 (§5.1).

R2 Edwards, W. (1977). “How to use multiattribute utility measurement for social decision making.” IEEE Transactions on Systems, Man, and Cybernetics, 7(5), 326–340.

USED FOR: SMART — the Simple Multi-Attribute Rating Technique — is precisely the method applied here: rate each attribute on a common anchored scale, weight, sum. Justifies the 0–10 rubric and direct weight allocation over more elaborate elicitation (§5.1, Step 4).

R3 Pöyhönen, M. & Hämäläinen, R. P. (2001). “On the convergence of multiattribute weighting methods.” European Journal of Operational Research, 129(3), 569–585.

USED FOR: Empirical evidence that different elicitation methods yield materially different weights. Underpins the caution in §12.1 that the weights are a prior requiring recalibration.

R4 Saaty, T. L. (1980). The Analytic Hierarchy Process. New York: McGraw-Hill.

USED FOR: The recommended upgrade path for weight derivation: pairwise comparison with a consistency ratio, giving an auditable defence of the weighting (§12.1). Also relevant to threshold derivation (§6.3).

B. Sales-force sizing and call planning

R5 Industry benchmark data on Indian pharmaceutical medical representative productivity: 8–12 field calls per day, varying by therapy area and territory density.

USED FOR: Validates the capacity assumption in Formula 3. The 8–10 range used sits at the conservative end of published practice, strengthening the feasibility claim (§7.1).

R6 Zoltners, A. A., Sinha, P. & Lorimer, S. E. (2004). Sales Force Design for Strategic Advantage. Basingstoke: Palgrave Macmillan.

USED FOR: The standard reference for the two accepted sizing methods — the workload (bottom-up) model used here, and the sales-response (ROI) model. Also establishes that launches require higher reach and frequency than mature-brand maintenance (§7).

R7 Lodish, L. M. (1971). “CALLPLAN: An interactive salesman’s call planning system.” Management Science, 18(4-II), 25–40. Extended by Manchanda, P., Rossi, P. E. & Chintagunta, P. K. (2004), “Response Modeling with Nonrandom Marketing-Mix Variables,” Journal of Marketing Research, 41(4), 467–478. doi:10.1509/jmkr.41.4.467.47005

USED FOR: The empirical foundation for differentiated call frequency: response to detailing is concave, so the constrained optimum equalises marginal return per call. This is the justification for the tier ladder derived in §7.3 — and the justification the source document omits. Manchanda et al. also warn that detailing is targeted non-randomly, biasing naive response estimates upward.

R8 Industry commentary on the limitations of response curves in biopharmaceutical call planning.

USED FOR: Supports the caution that response-curve estimation error is largest at high call frequencies, because few doctors are called that often — so the region determining Platinum frequency is the least reliable (§7.4).

C. Regulatory and compliance framework (India)

R9 Department of Pharmaceuticals, Ministry of Chemicals and Fertilizers, Government of India. Uniform Code for Pharmaceutical Marketing Practices (UCPMP) 2024, notified 12 March 2024, superseding UCPMP 2015; amended September 2025.

USED FOR: The binding constraint on every rupee in the budget model. Prohibits gifts, monetary grants, travel and hospitality to healthcare professionals (travel permitted only for CME speakers); caps free samples at 12 packs per drug per professional per year with total value $\leq 2\%$ of prior-year domestic sales; caps brand reminders at ₹1,000; prohibits foreign-location CMEs; requires Ethics Officers, Ethics Committees, annual CEO self-declaration and expenditure disclosure (§8.4).

R10 Apex Laboratories Pvt Ltd v. Deputy Commissioner of Income Tax, Supreme Court of India (2022); Explanation 3 to Section 37(1), Income-tax Act 1961; CBDT Circular No. 5/2012.

USED FOR: Establishes that non-compliant promotional spend is disallowed as business expenditure, making it doubly costly — the spend itself plus the lost deduction (§8.4).

D. Diffusion and adoption theory

R11 Bass, F. M. (1969). “A new product growth model for consumer durables.” *Management Science*, 15(5), 215–227.

USED FOR: Formula 8. Establishes that adoption follows an S-curve driven by innovation (p) and imitation (q) coefficients, providing the benchmark against which the plan’s linear ramp is assessed (§10.2–10.3).

R12 Rogers, E. M. (2003). *Diffusion of Innovations*, 5th edition. New York: Free Press. First edition 1962.

USED FOR: The theoretical basis for Pillar 2, Adoption Propensity. The canonical adopter categories — Innovators 2.5%, Early Adopters 13.5%, Early Majority 34%, Late Majority 34%, Laggards 16% — map onto the four-level rubric and supply the field-scoring audit test in §10.4.

R13 Sultan, F., Farley, J. U. & Lehmann, D. R. (1990). “A meta-analysis of applications of diffusion models.” *Journal of Marketing Research*, 27(1), 70–77.

USED FOR: Source of the benchmark diffusion parameters used to test the ramp shape: mean coefficient of innovation $p = 0.03$ and mean coefficient of imitation $q = 0.38$ across 213 applications (§10.2).

E. Financial appraisal

R14 Brealey, R. A., Myers, S. C. & Allen, F. (2020). *Principles of Corporate Finance*, 13th edition. New York: McGraw-Hill. Chapters 5–6.

USED FOR: The standard definitions applied in §12.4. Payback period is the time until cumulative net cash flow turns positive — the definition under which the Month 14–15 claim fails. Also the authority for supplementing payback with NPV, since payback ignores the time value of money.

R15 Horngren, C. T., Datar, S. M. & Rajan, M. V. *Cost Accounting: A Managerial Emphasis*, 16th edition. Harlow: Pearson. Standard definition: $ROI = (\text{gain} - \text{cost}) \div \text{cost}$.

USED FOR: The basis of the correction in §12.2: the source reports a revenue-to-cost ratio of 140% as “ROI”, where the correct figure is 40%, or approximately -2% once gross margin is applied.

F. Opinion leadership and peer influence

R16 Coleman, J. S., Katz, E. & Menzel, H. (1966). *Medical Innovation: A Diffusion Study*. Indianapolis: Bobbs-Merrill.

USED FOR: The foundational empirical study of prescription diffusion among physicians — the Columbia tetracycline study. Established that network-central doctors adopt earlier and that opinion-leader adoption precedes acceleration of the diffusion curve. The authority for Pillar 3. It also supplies an operational definition of “KOL”: three or more sociometric nominations — measured peer nomination rather than job title (§4.3, §12.2).

R17 Van den Bulte, C. & Lilien, G. L. (2001). “Medical Innovation revisited: Social contagion versus marketing effort.” *American Journal of Sociology*, 106(5), 1409–1435.

USED FOR: The essential counterpoint, and the reason the KOL business case must be rebuilt. Re-analysing the Coleman dataset with advertising-volume data, the authors found the contagion effect largely disappears once marketing effort is controlled for — the company had simply detailed influential doctors harder (§12.2).

R18 Nair, H. S., Manchanda, P. & Bhatia, T. (2010). “Asymmetric social interactions in physician prescription behavior: The role of opinion leaders.” *Journal of Marketing Research*, 47(5), 883–895.

USED FOR: A modern, econometrically identified estimate of opinion-leader influence: real, but considerably smaller than the 11.7× multiplier implied by the source document. Provides a credible range for rebuilding the KOL case (§12.2).

G. Concentration measurement

R19 Gini, C. (1912). *Variabilità e Mutabilità*. Bologna. Lorenz, M. O. (1905). “Methods of measuring the concentration of wealth.” *Publications of the American Statistical Association*, 9(70), 209–219.

USED FOR: Formula 7. The Lorenz curve and Gini coefficient convert the qualitative Pareto assertion into a measured statistic, providing the formal test of whether differentiated targeting is warranted at all (§10.1).

R20 Pareto, V. (1896). *Cours d'Économie Politique*. Lausanne: F. Rouge.

USED FOR: The origin of the concentration principle invoked throughout. Note that the source document's own tier table supports 70–80% of revenue from Platinum plus Gold, midpoint 75% — so the familiar “80%” should be presented as the top of a range rather than a point estimate (§2.2, §6).

H. Scheduling and territory design

R21 Graham, R. L. (1969). “Bounds on multiprocessing timing anomalies.” *SIAM Journal on Applied Mathematics*, 17(2), 416–429.

USED FOR: Formula 6. Territory balancing is makespan minimisation on identical parallel machines — NP-hard. The LPT greedy heuristic used carries a proven worst-case bound of $(4/3 - 1/3m)$ times optimal; at $m = 6$ territories this is 1.278× (§9.2).

R22 Zoltners, A. A. & Sinha, P. (2005). “Sales territory design: Thirty years of modeling and implementation.” *Marketing Science*, 24(3), 313–331.

USED FOR: Confirms that production territory design is a bi-criteria problem — balance workload and minimise travel — and that GIS-based optimisation is standard practice. The basis of the limitation acknowledged in §9.4.

I. Primary source

R23 Data-Driven Pharma Sales Planning & Doctor Targeting Model: Regional Sales Strategy Blueprint, Version 1.0, 51 pages, 20 sections.

USED FOR: The strategy document analysed in this report. All rubrics, tier definitions, budget figures, territory tables and financial projections are transcribed from it, with section references given throughout. All 123 of its numerical assertions were independently re-derived (§12.5).

Appendix A — Formula Reference Card

All formulas used in the model, with their section reference and supporting authority.

ID	Name	Expression	Section	Refs
F1	Composite Doctor Value Score	$DVS = \sum w_i \cdot s_i, \sum w_i = 1$	§5	R1, R2
F2	Tier classification	$T(d)=t \iff lo_t \leq DVS < hi_t$	§6	R4, R20
F3	Field capacity	$C = M \times D \times R$	§7	R5, R6
F4	Call-plan demand	$V = \sum n_t \cdot f_t; U = V/C$	§7	R7, R8
F5	Budget allocation	$B_t = B \times \alpha_t; b_t = B_t/n_t$	§8	R9, R10
F6	Territory balancing	$\min \max W_j$ (LPT heuristic)	§9	R21, R22
F7	Value concentration	$G = 1 - \sum (X_k - X_{k-1})(Y_k + Y_{k-1})$	§10	R19, R20
F8	Adoption dynamics	$n(t)=p[M-N]+q(N/M)[M-N]$	§10	R11, R13
F9	Marginal optimality	$g'_d(f_d) = \lambda$ for all funded d	§7.3	R7
F10	Balance measure	$CV = \sigma / \mu \leq 10\%$	§9.3	R22

Key parameters

Parameter	Value	Source
Pillar weights	0.30 / 0.25 / 0.20 / 0.15 / 0.10	Expert allocation (SMART); §4.2
DVS range	[1.75, 10.00]	Proved in §5.2
Tier thresholds	8.0 / 6.0 / 4.0	Policy choice; §6.3
Visit frequencies	3.5 / 2.0 / 1.0 / 0.33 per month	Derived from concavity; §7.3
Budget shares	45% / 28% / 15% / 12%	§8.1 (note the Bronze conflict, §8.3)
Field capacity	$6 \times 22 \times 9 = 1,188$ calls/month	§7.1
Bass parameters	$p \approx 0.03, q \approx 0.38$	Sultan et al. (1990); §10.2
LPT bound (m=6)	$1.278 \times$ optimal	Graham (1969); §9.2

DOCUMENT SUMMARY

This report documents a five-pillar multi-attribute scoring model for pharmaceutical doctor targeting under resource constraint. Ten formulas are derived from first principles, each with its theoretical authority, assumptions, limitations and a verified worked example. Twenty-three sources are cited and mapped to the specific formula they support. All 123 numerical assertions in the source strategy were independently re-derived; 85 verified exactly and six material errors were identified and corrected.